SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science & Engineering

CO1 - ASSESSMENT TOOL - 1

DESCRIPTIVE WRITTEN TEST

Course Code	CSA3717	Course Title	Software Testing
CO Assessed	CO1	Assessment	Assessment Tool 1: Descriptive Written Test
Weightage	50% of CIA	Total Marks	100
CO1 Statement	*Apply software testing principles to enhance the effectiveness of software quality assurance and improve defect detection strategies.* **(BL3)**		
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Section/Batch	Slot A	Date	21-10-2026

Code of Conduct:

I, **A.V.Rama krishna reddy** , certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student

Rama krishna

A. V. Rama krishna Reddy
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Structured Testing process for an ATM system.

Introduction: Software testing is a systematic process used to identify defects and ensure that software works according to requirements. An ATM system must be tested carefully because it handles money and customer account information. A structed testing process across all SDLC phases helps improve quality, security, and reliability.

Objectives: . to implement a structured testing process throughout the SDLC.

* To identify defects in ATM withdrawal functionally.
* To improve accuracy, security, and reliability.
* To reduce defects, through consistent testing practices.

Structured Testing process across SDLC phases.:

Required analysis: Identify ATM withdrawal requirements, such as pin verification, balance checking, withdrawel limits, and cash availability.

Test planning and Design: prepare a test plan, test cases, test data expected results, and testing schedules.

Unit testing: Test individual components, such as pin validation, balancing checking, and withdrawal amount calculation.

Integration Testing: test communication b/w the ATM, bank sever, account database, and cash dispenser.

System testing: Test the complete ATM withdrawal process, including valid PINs, insufficient balances, withdrawal limits, and network failures.

Acceptance Testing: Verify that the ATM meets customer and banking requirements before release.

Maintenance and Regression testing: After fixing defects or updating software, repeat relevant tests to ensure existing functions still work correctly.

Applying the Testing process to ATM withdrawal.

Test case	Input (or condition)	Expected result.
Valid withdrawal	Correct pin and sufficient balance	cash dispensed and balance updated.
Invalid PIN	Incorrect PIN	Transaction rejected.
Insufficient balance	withdrawal exceeds available balance	Transaction rejected without deduction.
withdrawal limit	Amount exceeds daily limit	Transaction rejected.
cash unavailable	ATM has insufficient cash	No deduction; clear message displayed.
Network failure	Connection lost during transaction	Transaction handled safely without duplication deduction
Invalid amount	Zero or negative amount	request rejected.
Successful Transaction	valid withdrawal completed.	Receipt or confirmation provided.

How process Consistency Reduces Defects.

* Standard procedure : Ensure all testers follow the same testing steps.

* Complete Test Coverage : Reduce the chance of missing important scenarios.

* Early detection: Identify defects before they reach production.

* Defect documents : Help developers reproduce and fix problems

* Regression testing : Ensure fixes do not break existing functionality.

* Team collaboration : Impove communication between testers, developers, and banking teams.

Expected outcomes :

* Accurate ATM withdrawels and account balance updates.
* fewer transaction failures and duplicate deductions.
* Improved security and customer confidence.
* Reduced defect Correction Costs.
* Reliable performance under normal and exceptional conditions.
* Better software quality through repeatable testing practices.

Conclusion :

A structured testing process across all SDLC phases helps identify defects early and ensures that ATM withdrawel functionally works correctly. Consistant test planning, execution, defect tracking, and regression testing reduce failures and improve reliability. As a result, customers receive a safer, more accurate, and dependable ATM service.

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions	
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation	
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis	
Organization	20%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers	
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps	